

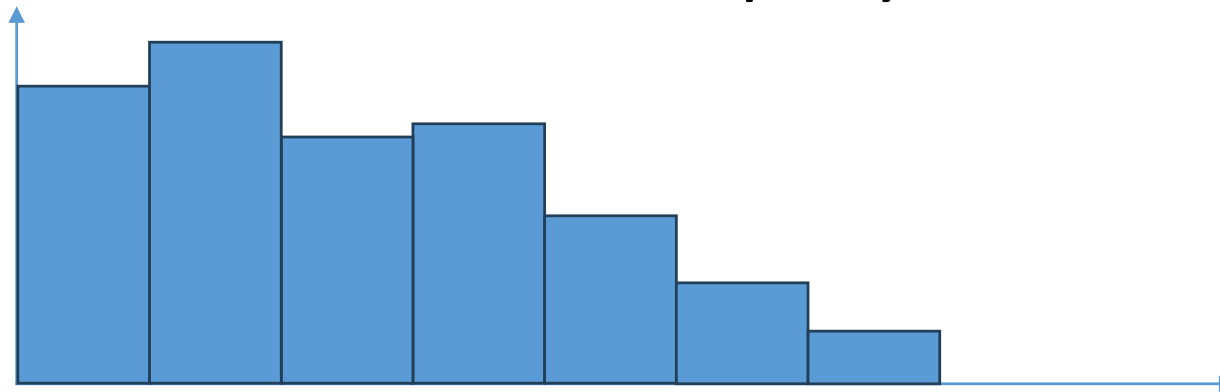
Empirical vs Theoretical Distribution

Empirical distribution

- Derived **directly from observed data** in a sample, describes **how the observed values are distributed**
- Depends on the **specific sample** collected
- Can be represented using:
 - histogram
 - relative frequencies

Example:

- observed *execution times* of an activity: $t_1, t_2, \dots, t_n$
- The empirical distribution describes **how frequently** the observed values occur in the sample.



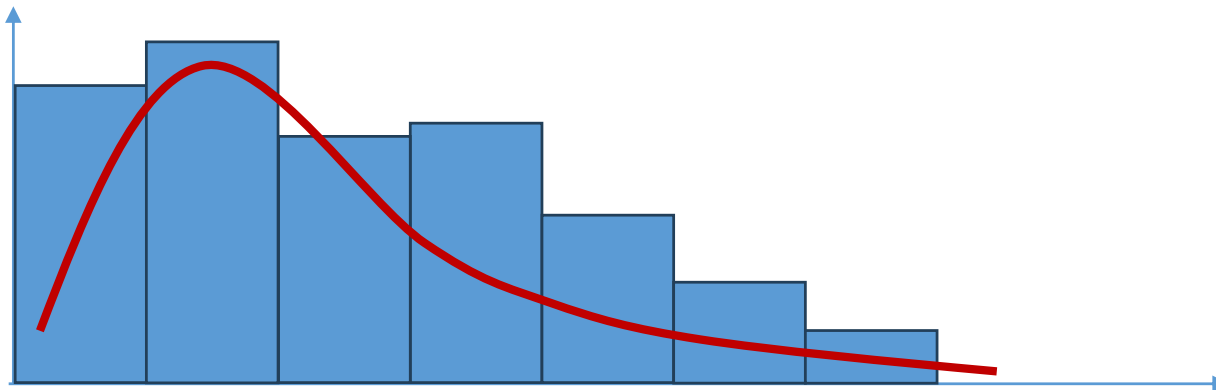
Empirical vs Theoretical Distribution

Theoretical distribution

- A **mathematical probabilistic model**
- Describes the **expected probabilistic behavior** of a random variable
- Does not come directly from the data but from **assumptions about the phenomenon**

Example:

- $f(x) = \lambda e^{-\lambda x}$ $x \geq 0$ (exponential distribution)



Empirical are used to **estimate the parameters of the model**

$$\mu = \frac{1}{\lambda} = \frac{1}{n} \sum_{i=1}^n t_i$$

Acquire and Analyze and model data

- Once a set of samples representing a system data input has been acquired,
- there are three choices for generating the random variate stream required for simulation:
 - the sample values can be used directly to form the random variate stream.
in a limited set of sample values, legal and realistic values that **occur infrequently** may not be present
 - the set of samples can be used to generate an empirical distribution and the empirical distribution in turn can be used to generate the random variate stream.
the advantage of facilitating the generation a random variate stream of **unlimited length**. It cannot generate **outlying values**.
 - Third, a theoretic distribution function that fits the set of samples can be identified, and this theoretical distribution can be used to
The theoretical distribution will generate values **outside the range of observed Values**. Also, it **smooths irregularities**.

DESIGN DETAILED EXECUTABLE MODEL

- Use of a *general-purpose programming language*
 - C, C++, Java
- Use of a *simulation-specific language*
 - SIMULA, GPSS, JavaSim
- Use of a *simulation tool*
 - Arena, ExtendSim, Simul8



VERIFY & VALIDATE MODEL

- Why?



- Is Cavity Sam suitable for medical training?



Simquest Sim

VERIFY & VALIDATE MODEL

- Why?

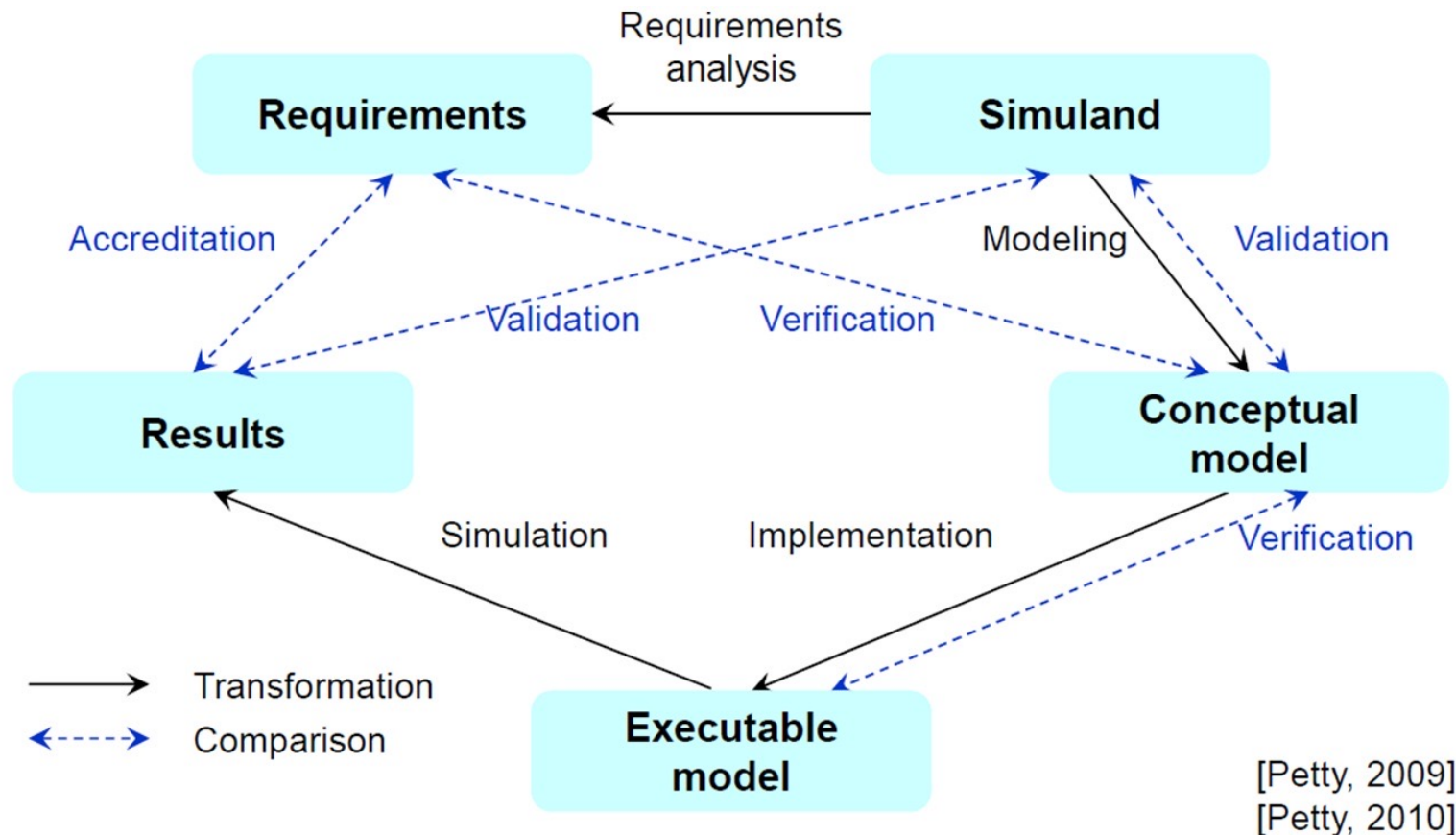


- Is Zero-flight time simulation suitable for pilot training?

Verify & Validate Model

- Make our M&S assets more credible and reliable
- Verification
 - Determine that model implementation and data accurately represent conceptual description and specification
 - Is the model implemented right?
- Validation
 - Determine degree to which the model and data are accurate representation of the real world from the perspective of its use.
 - Is the right model implemented?

Verify & Validate Model



Verify & Validate Model

Informal	Static	Dynamic	Formal
-Audit -Desk checking -Documentation Checking -Face validation -Inspections -Reviews -Turing test -Walkthroughs	-Cause-Effect Graphing -Control Analysis -Data Analysis -Fault/Failure Analysis -Interface Analysis -Semantic Analysis -Structural Analysis -Symbolic Evaluation -Syntax Analysis ...	-Acceptance Testing -Alpha Testing -Assertion Checking -Beta Testing -Bottom-up Testing -Comparison Testing -Statistical Techniques -Structural Testing -Submodel/Module Testing ...	-Induction -Inductive Assertions -Inference -Logical Deduction -Lambda Calculus -Predicate Calculus -Predicate Transformation -Proof of Correctness

[Balci, 1998]

Experiment Design

- Empirical methods for proving hypothesis or comparing alternative approaches
- Independent Variables - manipulated by Experimenter
- Dependent Variable – measured by Experimenter

Experiment Design

- Select measures of performance
 - Example: aircraft losses, cost, time
- Select factoring parameters
 - Example: aircraft loads, technical specifications, courses of actions
- Decide about value range and steps for controlled variables
- Decide how to factor
 - Example: Full factoring, partial factoring
- Decide the technique to build confidence interval
 - Example: Single run with multiple intervals, multiple runs with a single interval

Experiment Design

- Value ranges and steps
 - Aircraft loads (5 different load)
 - Weather Conditions (Open, windy, rainy, snow, windy, thunderstorm, hurricane)
 - Number of personnel (Between 100 and 200 with step of 10s)
- Factoring
 - Full Factoring ($5 \times 7 \times 10 = 350$ experiment configurations)
 - Partial Factoring (For example, use only the best configuration of weather and personnel numbers for testing aircraft loads ($7 \times 10 + 5 = 75$ experiment configurations))

EXECUTE SIMULATION

- Run/Execute Simulation
- Collect data (DB)
- Analyze output data

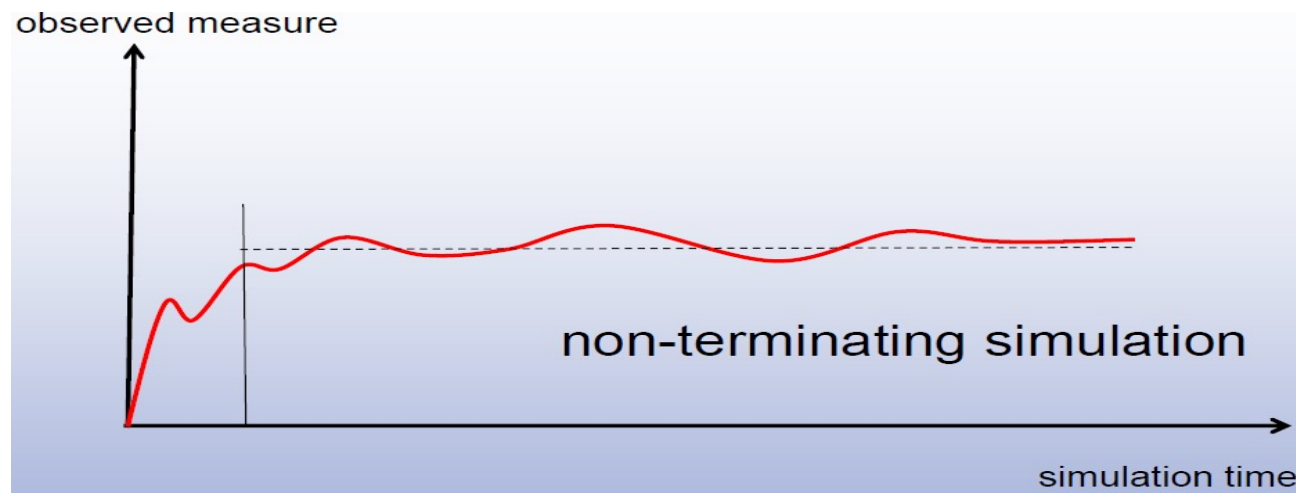


ANALYZE OUTPUT DATA

- Draw conclusions and make recommendations
- Operations Research
- Visualization – displaying and plotting (2D, 3D)
- Random Inputs → different outputs – statistics (estimation)

ANALYZE OUTPUT DATA

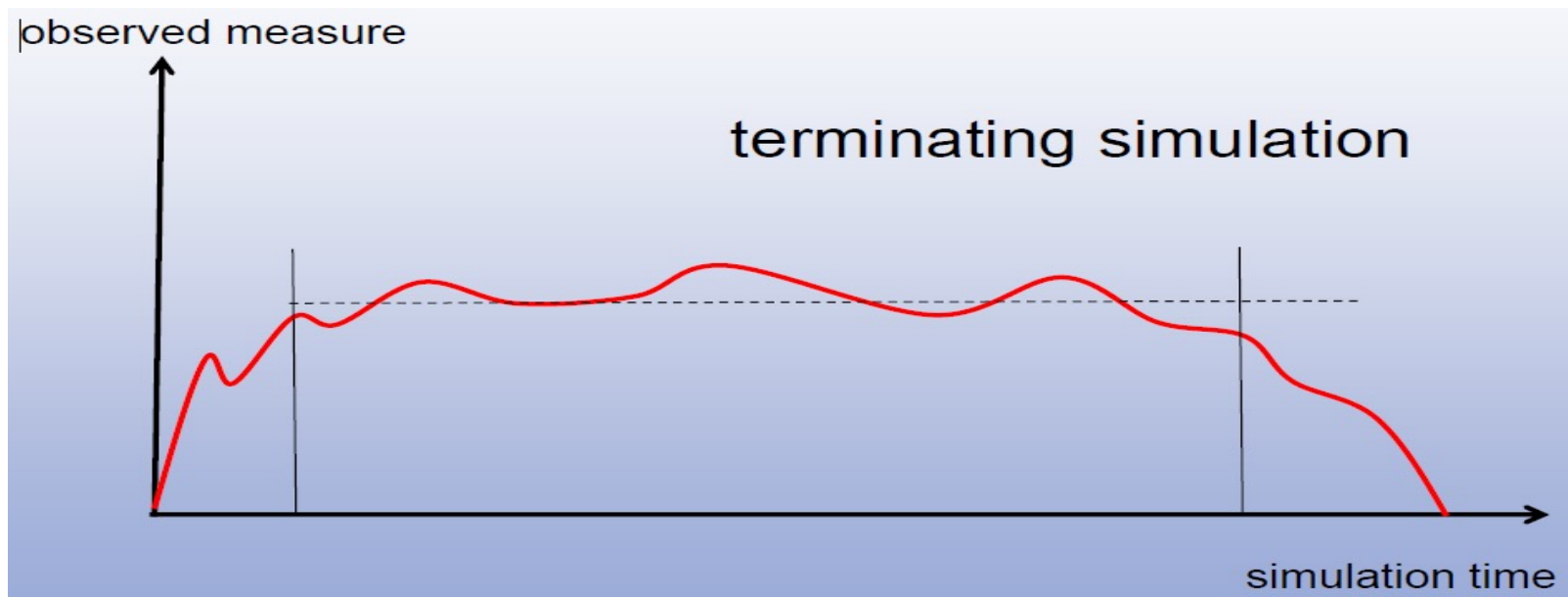
- Terminating or non-terminating simulation
- Initial Transient :Warming up period or initial data deletion



- Replication/Deletion approach: Replicate the simulation n times and delete the warm up period (l) from each, then construct confidence interval

ANALYZE OUTPUT DATA

- Length of non-terminating simulation
- Cool Down period for terminating simulations



*the most critical, skills-requiring
and error-prone activity*

Activities in a sound M&S study

